MicroBooNE NuMI FHC charged-current $\mathbf{1\mu\,1\pi^{\pm}\,Xp}$ differential cross sections: the XSecAnalyzer framework

Framework-only summary note

July 25, 2026

This note documents the *XSecAnalyzer* framework extraction of the MicroBooNE NuMI FHC charged-current ν_μ $1\mu\,1\pi^{\pm}\,Xp$ differential cross section. It describes the inputs, the method, and the results of the framework pipeline alone; the parallel custom-macro pipeline and the cross-pipeline comparison are documented elsewhere.

1 Inputs

The measurement uses the MicroBooNE **Run 1 NuMI FHC** samples. The beam-on data slot is filled with **NuWro fake data** for a generator-closure test (beam-on data is disabled). The simulated samples are:

- GENIE ν_μ overlay (`numuMC`), which defines the detector response (smearing) and efficiency;
- beam-off (EXT) data for the cosmic background;
- the dirt (out-of-cryostat) sample;
- eight detector-variation samples for the detector systematic uncertainty.

Normalisations: data POT 3.283×10^{20}; the simulation is scaled by 0.14101 (run POT / summed MC POT); the NuWro fake-data slot corresponds to 6.65042×10^{20} POT. The NuMI $\nu_\mu + \bar{\nu}_\mu$ flux is integrated to $\Phi = 2.372512 \times 10^{-9}$ cm^{-2} POT^{-1}, and the argon target count is taken over the fiducial volume with the z dead region (675.1–775.1 cm) excluded. The configuration is driven by `file_properties_numi.txt`, the per-observable `ccpi_<obs>_bin_config.txt`, `ccpi_xsec_config_numi.txt`, and `ccpi_systcalc_numi.conf`.

2 Method

Signal definition. A signal event is a charged-current ν_μ interaction with a true vertex in the fiducial volume, exactly one muon and exactly one charged pion, any number of protons (Xp, $X \geq 0$), and no other mesons. The cross section is reported in the **measured phase space**

$$p_\mu > 0.15\ \text{GeV}/c, \qquad p_\pi > 0.175\ \text{GeV}/c, \qquad \theta_{\mu\pi} < 2.6\ \text{rad}, \tag{1}$$

which removes the low-momentum and large-opening-angle regions where the selection has little acceptance.

Selection. The `CC1mu1piXp` selection requires a reconstructed neutrino vertex in the fiducial volume with topological score > 0.67 and at least two tracks; it identifies the muon candidate as the longest qualifying track (muon PID) and the pion candidate as the highest-LLR qualifying track, applies track-containment, gap, and shower-veto requirements, the opening-angle cut $\theta_{\mu\pi} < 2.6$, and an upper track-multiplicity bound. The muon momentum is reconstructed from multiple Coulomb scattering (`candidate_muon_mom_mcs`) and the pion momentum from the proton-hypothesis range.

Extraction chain. The processed ntuples are produced by `ProcessNTuples`; `univmake` then builds the response matrix, backgrounds, and the full set of systematic universes per observable; and `Unfolder` performs a **Wiener–SVD** unfolding (second-derivative regularisation) using the full systematic-plus-statistical covariance. The cross section in each true bin is obtained from the unfolded signal N via the conversion factor,

$$\sigma = \frac{N}{N_{\mathrm{Ar}}\,\Phi}, \tag{2}$$

with N_{Ar} the in-FV argon target count. The systematic budget includes the flux (PPFX), GENIE cross-section (UBGenie and unisim/reweight), detector variations, hadron reinteraction, dirt normalisation, finite MC/EXT statistics, and POT.

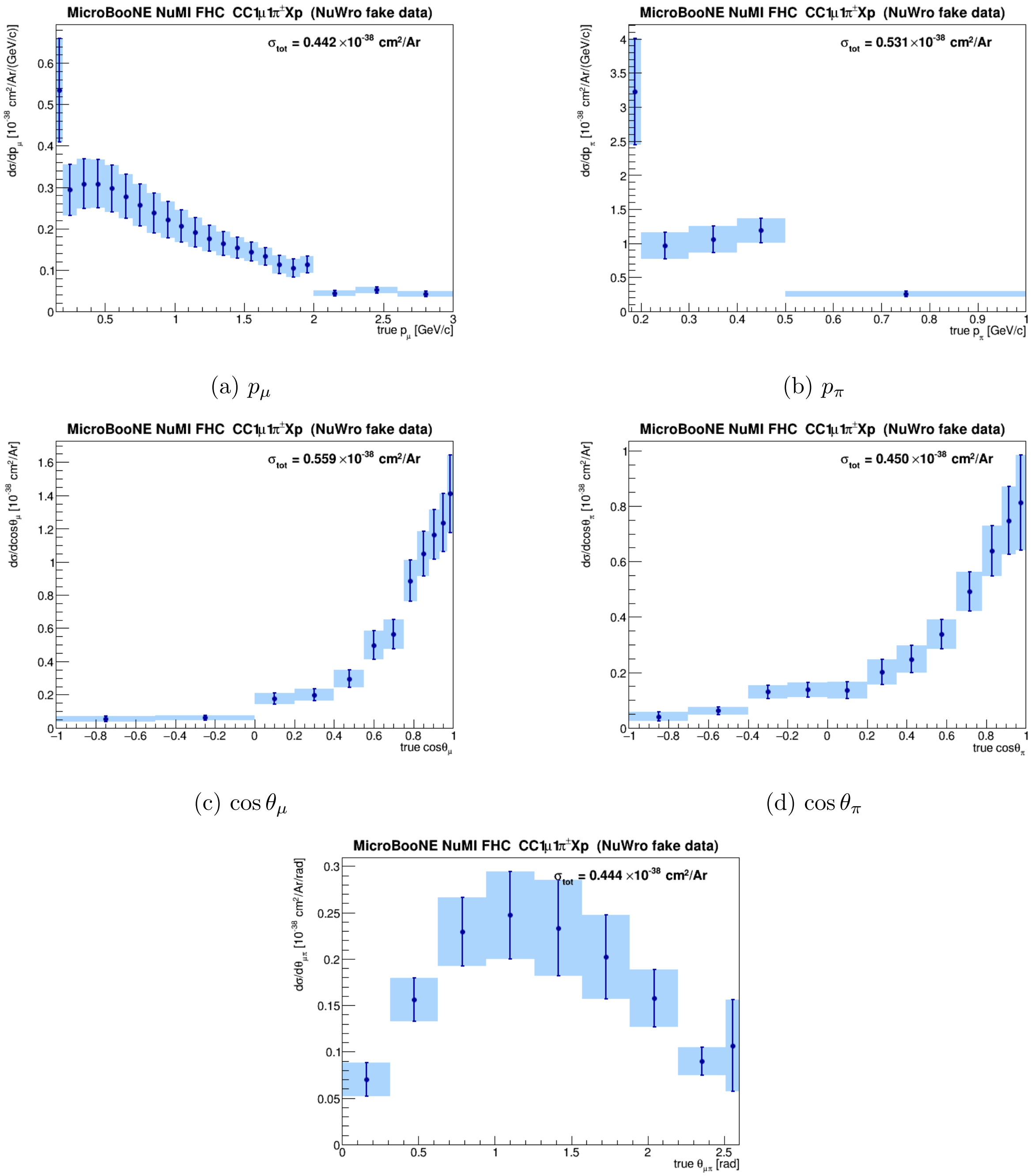
Validation. The framework passes the NuWro fake-data closure test: unfolding the NuWro pseudo-data through the GENIE-built response recovers the input generator truth within the quoted statistical precision, confirming that the response, background subtraction, and unfolding are internally consistent.

3 Results

Table 1 lists the integrated cross section obtained for each observable in the measured phase space, and Fig. 1 shows the corresponding differential distributions with their full (systematic plus statistical) uncertainty bands.

Table 1: Integrated framework cross section per observable (measured phase space), in units of 10^{-38} cm^2/Ar.

Observable	σ [10^{-38} cm^2/Ar]
p_μ	0.442
p_π	0.531
$\cos\theta_\mu$	0.559
$\cos\theta_\pi$	0.450
$\theta_{\mu\pi}$	0.444



(a) p_μ

(b) p_π

(c) $\cos\theta_\mu$

(d) $\cos\theta_\pi$

(e) $\theta_{\mu\pi}$

Figure 1: Framework Wiener–SVD unfolded differential cross sections $d\sigma/dx$ (NuWro fake data) in the measured phase space. Points are the unfolded result; the shaded band is the full systematic-plus-statistical uncertainty. The first p_μ/p_π bins and the last $\theta_{\mu\pi}$ bin are narrow because they are truncated at the phase-space boundaries.